The Distributional Effects of Wildfires on House Prices in the Western United States

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May, 2021

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Abstract

This paper analyzes the effects of wildfires on house price in the western United States. Results from this analysis show a small, but significant negative impact of wildfires on house value when effects are analyzed on a zip code level. A -0.85% change is found in a zip code's average house value when a zip code shifts from the 25th percentile in fire instances to the 75th percentile in fire instances, and pre-fire trends are considered. A larger effect of wildfires on house price is found in states with a lower average house value, and a larger effect is found in 25th quantile of house price than in the 75th. There is also evidence that vegetation amplifies the negative impact of wildfires on house price, and increased land development dampens the effect, though this is not consistent across states. The large average size of zip codes in the western US may cause for the magnitude of the effect of wildfires on house price to be greatly underestimated.

Introduction

It is clear that as the amount and severity of wildfires has steadily increased due to climate change in the past 20 years (Hessl 2011, Marlon et al. 2008, Westerling & Bryant 2007). While the impact of wildfires on vegetation, ecology, and infrastructure is well known, the economic impacts of increased wildfires is still an area of active study. As wildfire presence is increasing, it is important to understand the impact of wildfires on house price. House prices in particular have been studied as an indicator of the impact of wildfires on a community, as change in average house price reflects the market's assessment of the value of living in a particular area. The Hedonic Property Model (HPM) is the primary theoretical framework which models the relationship between a property's characteristics and its price (Huggett, Murphy, & Holmes 2008). The HPM is a popular method to explain the positive effect of amenities, such as

woodlands, on house prices (Dombrow *et al.* 2000). Disamenities can also be analyzed through the HPM, as the presence of frequent floods, earthquakes, and poor air quality has been shown to lower the value of a house (Bin & Polasky 2004, Bernknopf *et al.*1990, Liu *et al*. 2018). In addition to actual present disamenities, the perceived risk of damage to amenities or increasing disamenities, affects a house's price. Thus, if a homeowner or homebuyer perceives a risk of wildfires repeatedly affecting a home, air quality, or nearby amenities, the Hedonic Property Model reflects that the market valuation of the home will reflect this.

This paper studies the impact of wildfires on house prices in the western United States, and in particular, whether the effect is consistent across states, and if the effect is amplified by the vegetation and development present in the area. The western US was chosen as the area of study as there has been consistent increase in wildfires across the region (Westerling 2016). Additionally, there is a wide range of vegetation type and density in the western US whose impact can be studied.

Most of the past studies into the effects of wildfires on house prices focus on a particular fire or limited geographic area. Loomis (2004) studied the Buffalo Creek fire in Pine, CO and found that house prices decreased by 15% following the fire. Donovan et al. (2007) find that the publication of a website that rated fire risk in Colorado Springs had a significant negative impact on housing prices. This shows that an increase in perceived risk alone can impact house price. Similarly, Kiel & Matheson (2018) studied the price of "at risk" homes while excluding homes directly impacted by the 2010 Fourmile Canyon Fire in the Boulder, CO area, and find a substantial negative impact on price. These studies clearly show that perceived risk of wildfire damage is revealed through a drop in house price.

In contrast, Baylis and Boomhower (2019) found that large public expenditures on firefighting efforts inadvertently subsidize expansion of housing development into areas of high risk for wildfires. As such, large expenditure on firefighting efforts can reduce the impact of wildfires on house value. If fires are no longer seen as a risk to houses and nearby amenities, house prices may not respond to an increase in nearby wildfires. This brings into question whether a negative effect from wildfires on house price can still be seen in all areas that are at risk for wildfire.

Vegetation and forest presence have been shown to impact house prices in areas that have undergone repeated wildfires. Roman, Azqueta & Rodriguez (2013) show that vulnerability to wildfires is greatly influenced by the present carbon stock and buildings. This implies that higher vegetation presence should increase the perceived risk of wildfires causing damage to houses, and thus lower a zip code's average house value. Loomis (2004) shows that house value decreases if nearby forests are destroyed by wildfire, as the amenity value is decreased. As such, it seems likely that high vegetation density will amplify the negative effect of wildfires on house price.

The analysis in this paper is most closely based on Mueller, Loomis & González-Cabán's 2007 paper on the effect of repeated wildfires on residential property values in Southern California. Loomis finds that repeated forest fires cause house prices to decrease, and the first fire in an area has a smaller affect on house price than the presence of a second fire. Mueller and Loomis' 2014 paper extends this study to examine the effect across the price distribution, using a quantile regression approach. They find a large difference between the impact on the 75th quantile and the 25th quantile, with a larger impact on the 75th quantile. This paper examines if this result holds on a larger scale. This paper studies zip codes in 10 states to see if the presence

of repeated wildfires in a zip code has a consistent effect on house price, or if it can be seen to differ with vegetation and along the price distribution.

Results from this analysis show a small, but significant negative impact of wildfires on house value is consistent across the western United States. A -0.44% change is found in a zip code's average house value when a zip code shifts from the 25th percentile in fire instances to the 75th percentile in fire instances. When pre-fire differences are in average house value are considered a slightly larger effect, -0.85%, is found. Additionally, this effect is found to be stronger on homes of lower value. A larger effect of wildfires on house price is found in states with a lower average house value, and a larger effect is found in 25th quantile of house price than in the 75th. There is also evidence that vegetation amplifies the negative impact of wildfires on house price, and increased land development dampens the effect, though this is not consistent across states. The large average size of zip codes in the western US may cause for the magnitude of the effect of wildfires to be underestimated, as fires may not actually affect the price of many houses if a zip code is large. This suggests that future should research use smaller units of area to study the impact of wildfires on house price to account for more precise spatial impacts.

Data

To estimate the price of affected homes, this paper uses a typical house value estimate from Zillow, an online real estate market company. The smallest unit of data for which geographic coordinates and house value data was available is at zip code level. The total market valuation for a zip code is determined by a weighted model of each home's valuation. Each home is weighted by its valuation divided by the total sum of all homes valuation in the market.

A weighted model is used so no outliers can skew the estimated typical value of a home in a zip code. The total market valuation the is the sum of each home's weighted value.

The zip code typical house value is then seasonally adjusted, and smoothed by a three month moving average. Using an estimated typical house value as a proxy for home price is limited by the accuracy of the valuation provided, however Zillow reports a 1.9% median error rate for on-market homes, and a 7.5% median error rate for off-market homes and 95% of its estimates are within 10% of actual transaction price. These house values were then put into real terms, using the US Bureau of Labor Statistics' consumer price index with a 1982 base year.

Data on wildfires was gathered from NASA's Fire Information Resource Management System (FIRMS). The MODIS (Moderate Resolution Imaging Spectroradiometer) aboard the Terra and Aqua satellites capture fire data for the globe, and provide at least twice daily observations for all areas. All fire observations for the western united states from 2001-2020 were collected. Each fire observation represents an "instance" of fire, meaning that the MODIS registered a fire centered at a particular location at a certain time. Fires must be at approximately 1000 meters squared at least to be registered by MODIS, thus only significantly large fires are counted. Additionally, the fires are only counted only if they are categorized by FIRMS as a wildfire, and not as an industrial or housefire. The fire observations were matched with a zip code by determining the closest zip code centroid to the centroid of the fire, using the Euclidean norm as follows:

$$min_{i,j}\ (\sqrt{(latitude\ zipcode_i - \ latitude\ fire_j)^2}$$

$$+ \sqrt{(longitude\ zipcode_i - longitide\ fire_j)^2}\ )$$

While this does lead to some error of the exact zip code in which a fire is located within; as risk of fire is the main variable of interest, it is more important to determine proximity of a fire than the exact zip code of the fire. Then for each zip code the number of fires instances recognized per month was tabulated.

Vegetation data was gathered from the National Landcover Database for 10 states, Arizona, California, Colorado, Idaho, New Mexico, Nevada, Oregon, Texas, and Utah. The latest year of data released by the National Landcover Database is for 2011, which is halfway through the 19 years analyzed. For each zip code in these states, the reported coverage (in 30x30 square meters resolution) of 10 types of vegetation were added and then divided by the total area of the zip code to create a percentage of vegetation for each zip code. In addition, the amount of forest cover tabulated in the same manner. From the National Landcover database, data on the amount of developed land per square mile of each zip code is also available in 2011. Here land development refers to the area covered with buildings or roads per square mile. The percentage of highly developed land, partially developed land, and minimally developed land was also calculated. Below descriptive statistics for the entire data set are displayed, with standard deviation of each statistic in parentheses, and descriptive statistics for each are displayed in Appendix Tables 3 and 4.

Table 1: **Summary Statistics**

	All Zip Codes	Zip Codes with Fires	Zip Codes with No Fires
Total of Zip Codes Analyzed	2,890	1,576	1,313
Mean Zip Code Area Area (Square Mile)	84.5 (238.8)	186.2 (384.9)	57.5 (171.2)
Mean Annual Fire Instances per Zip Code		2.4 (22.3)	
Mean Monthly Fire Instances per Square Mile	0.00903 (0.981)	0.25 (5.16)	
Mean House Value	276,387 (229,391)	291,880 (231,102)	218,042 (143,322)
Mean %Vegetation Cover	48.5 (36.9)	79.2 (24.2)	40.3 (35.3)
Mean %Development	46.7 (37.3)	16.8 (23.2)	54.7 (36.3)

Standard deviation in parentheses

While this dataset is not ideal, it does provide a large sample of zip codes which have had multiple wildfires in the past 19 years. Zip codes analyzed are on average 84.5 square miles, with a high standard deviation of 239 miles, and as such some zip codes may yield better analysis than other. This provides a major limitation for what can be learned from the analysis of house prices on a zip code level. If a zip code is large, fires in one part of a zip code may not have an impact on the value of many homes within a zip code. Thus it seems likely that this paper underestimates the impact of fires on house price, as fires may be counted as present in a zip code where houses are not actually affected, and thus their prices do not respond to fire presence.

Methodology

First, the monthly house value is regressed on the count of wildfire instances per month for each zip code for all 10 states, over 19 years. The effect is lagged monthly to determine if the effect changes or remains constant over time. Zip code level fixed effects are used to absorb

differences in house price that are not cause by wildfire presence. Month by year dummies are used to absorb changes in house price over time that are not due to wildfire presence. The log of house value is used as the main dependent variable to allow the marginal effect of each independent variable to vary with the typical house value of a zip code, which is modeled after the methodology in Mueller, Loomis & González-Cabán's 2007 paper. The log-linear regression equation is:

$$log(HouseValue_{i,t}) = \beta_0 + \beta_1 FireCount_{i,t} + \beta_2 FireCount_{i,t-1} + \beta_3 FireCount_{i,t-2} + \cdots + \beta_{12} FireCount_{i,t-5} + \alpha_i + \delta_t + e_{i,t}$$

Here α_i represents zip code fixed effects and δ_t represents month by year dummy variables. Standard errors in this model are clustered to allow for error to vary by zip code. Additionally, this model was used to look at the effects for each state individually.

To reveal how the effects differ across the price distribution, a quantile regression approach is used. Here the effects of the fire instances are functions of the quantile of house price being analyzed.

$$Q_q(HouseValue_{i,t}) = \beta_0(q) + \beta_1(q) FireCount_{i,t} + \alpha_i + \delta_t + e_{i,t}$$

Here q is the desired quantile, which was set to 0.25, 0.5 and 0.75, α_i represents zip code fixed effects and δ_t represents month by year dummy variables. This model was also used for each state individually.

To find if vegetation density has an amplifying or mitigating impact on the effect of wildfires on house value, the percentage of vegetation was interacted with the fire count for each zip code. Thus for each zip code vegetation is allowed to vary such that the combined effect of vegetation and increased fires can be seen:

$$log(HouseValue_i) = \beta_0 + \beta_1 FireCount_i + \beta_3(FireCount_i * Vegetation\%_i) + \alpha_i + \delta_t + e_i$$

Similarly, the percentage of forest coverage was substituted for vegetation to see if the specific presence of forest cover differs from the presence of general vegetation. To find the combined effect of developed land and wildfire presence, developed land percentage was also interacted with fire count:

$$log(HouseValue_i) = \beta_0 + \beta_1 FireCount_i + \beta_3(FireCount_i * Development\%_i) + \alpha_i + \delta_t + e_i$$

The effect of development was tested for four levels of development, minimally, moderately, highly and all developed land.

To account for the difference in house value between zip codes with wildfires and zip codes with no wildfires, a one month lead on the effect of wildfires on house value was calculated. In this sample, zip codes with no fires have a higher average house value than zip codes with fires. This may cause an underestimation of the effect of wildfires on house value. The effect of fires on the previous month's house value was subtracted from the estimated effect of fires on house value.

$$log(HouseValue_{i,t}) = \beta_0 + \beta_1 FireCount_{i,t} + \alpha_i + \delta_t + e_{i,t}$$

$$log(HouseValue_{i,t}) = \theta_0 + \theta_1 FireCount_{i,t+1} + \alpha_i + \delta_t + e_{i,t}$$

$$Total\ Effect = \beta_1 - \theta_1$$

Although the average zip code size is large in the western United States, 84.5 square miles, some spillover effects of wildfires on nearby zip codes may occur. Fire instances are matched by least distance between the centroid of a fire instance and the centroid of a zip code. As such, fires may actually occur in a different zip code from which they are reported, or may be

near the border between two zip codes. These fires may then affect the average house value in more than one zip code. To capture these spillover effects, the amount of fires with second-least distance to a zip code were tabulated. These second-least distance fires are recorded as being closer to the centroid of another zip code, and are second closest to the centroid of the zip code for which they are recorded as "second-least distance."

Second Least Distance Calculation:

$$mi\,n_{i \neq least\ distance\ zipcode,j} \left(\sqrt{(latitude\ zipcode_i\ -\ latitude\ fire_j)^2} + \sqrt{\left(longitude\ zipcode_i - longitide\ fire_j\right)^2} \right)$$

Then the amount of fire instances with least distance and second-least distance to a certain zip code were summed to see the effects of all nearby fires on a zip code's average house value. The same regression analysis was performed on the sum of the least distance and second-least distance fires. The amount of third-least distance fire instances was also tabulated, but no significant effects were found from third-least distance fires on a zip code's average house value.

Results

When analyzing the effect of wildfires on house value by zip code for the entire dataset, no significant effect was found. However once zip codes from California are removed from the data set, a significant negative effect of wildfires on house value is found. In California, no significant impact of wildfires on house price is found. This may be because the demand for housing in California is so high that it homebuyers are willing to move into high risk areas. Thus excluding California, an additional fire per square mile in a zip code leads to a -0.0863% change in the house value. This represents a -0.44% change in a zip code's average house value when a

zip code shifts from the 25th percentile in fire instances to the 75th percentile in fire instances, for a zip code of average area. This result is in line with pervious literature's finding that there is a negative impact of wildfire fire presence on house value, however it is much smaller in magnitude. A -0.44% change represents a $1,284 change when the average house value ($291,880) is used, which is a somewhat negligible amount. This provides evidence that analysis at the zip code level greatly underestimates the effect of fires on nearby houses. These results are shown in Table 2:

Table 2: **OLS Model, Effect of Fire Instances on House Value**

	Log(House Value)	Log(House Value)	Log(House Value) Excluding CA	Log(House Value) Excluding CA
Fire Count/sqmi	-0.000291	-0.000293	-0.000836***	-0.000704***
	(0.000262)	(0.000219)	(0.0000960)	(0.0000747)
1 Month Lag		-0.000243		-0.000490***
		(0.000177)		(0.0000336)
2 Months Lag		-0.000238		-0.000523***
		(0.000190)		(0.0000456)
3 Months Lag		-0.000243		-0.000585***
		(0.000208)		(0.0000371)
4 Months Lag		-0.000255		-0.000576***
		(0.000208)		(0.0000387)
5 Months Lag		-0.000267		-0.000572***
		(0.000213)		(0.0000276)
6 Months Lag		-0.000305		-0.000856***
		(0.000282)		(0.0000709)
Observations	658,920	641,580	403,332	392,718
Adjusted R^2	0.969	0.969	0.958	0.958
Zip Code Fixed Effects	Yes	Yes	Yes	Yes
Month x Year Dummies	Yes	Yes	Yes	Yes

Clustered standard errors in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

The small negative impact of fire on house value is found to be somewhat permanent. The effect of a fire remains relatively constant on future months' house value, with some fluctuation in magnitude. This may be because the physical damage of a fire on infrastructure and natural amenities is seen as permanent and decreases house value, or the perceived risk of

fire is increased permanently after a nearby fire occurs. The effect remained fairly constant and significant for 15 months, and then significance of the result tapers off.

When the pre-fire difference in average house value between zip codes with fires and without fires is considered, a larger effect is found. An additional fire per square mile in a zip code leads to a -0.167% change in the house value, when the pre-fire difference in average house value is considered. This represents a -0.85% change in a zip code's average house value when a zip code shifts from the 25th percentile in fire instances to the 75th percentile in fire instances, for a zip code of average area. This result is approximately twice as large as the effect when pre-fire differences are not considered, but is still very small. These results are displayed in Appendix Table 6.

In order to better understand the above results, analysis on each state was done to try to find an explanation for the significance of fires in some states and not others. A negative effect of wildfires on house value was found in all states except California, Idaho, and New Mexico. As discussed above, the housing markets in California are not significantly impacted potentially due to demand for housing in California overshadowing negative impacts of present wildfires. Idaho and New Mexico may fail to show negative results due to their relatively small sample sizes, only 21 zip codes in Idaho and 26 zip codes in New Mexico with fires are analyzed. The largest magnitude of effect was found in Texas, which has an -8.1% change in house value from an additional fire per square mile. This represents a -11% change in house value when fires instances in a zip code increase from the 25th percentile to the 75th percentile, for the average sized zip code in Texas. When pre-fire differences in average house value are accounted for, a -16.23% change in house value from an additional wildfire per square mile is found, and a -25.7% change in average house value is found when fires instances in a zip code increase from the 25th

percentile to the 75th percentile, for the average sized zip code in Texas. In contrast to California which has the highest average home value, Texas has the lowest average home value of the states analyzed. This may signify that in Texas, homes have lower quality and more prone to damage by fire. These results are displayed in Appendix Table 5.

Next, the quantile model was used to see if the impact of wildfires on house value changes along the price distribution. When California zip codes are excluded from the data set, the largest magnitude of impact was found in the 25th quantile, which had a -$125 impact of an additional fire per square mile on house value, and the lowest magnitude was found in the 75th quantile, which had a -$100 impact of an additional fire per square mile. This result appears to contradict Loomis and Mueller's 2014 study in Southern California which found a greater effect on higher valued houses than lower valued homes. However, when the quantile regression was run on California alone, no significant effects were found, however, the largest magnitude of effect was found in the 75th quantile. This provides further evidence that the effect of wildfires in California differs from other states, and zip codes in California should be analyzed separately. These results are shown in Appendix Tables 7 and 8.

The vegetation and forest coverage in a zip code amplifies the negative effect of wildfires on house value. When a zip code's fire instances increase from the 25th percentile to the 75th percentile, a 1% increase in vegetation density leads to an additional -0.012% impact of wildfire on house value. This effect is larger when only forest coverage is considered instead of all vegetation types. When a zip code's fire instances increase from the 25th percentile to the 75th percentile, a 1% increase in forest coverage leads to an additional -0.028% impact of wildfire on house value. Forest presence may be seen as an amenity, whereas vegetation presence in general is not, and thus forest destruction from wildfires has a higher impact on house value. When the

impact of forest presence is analyzed for each state individually, the largest significant impact was found to be in Oregon, which has the highest average percentage of forest coverage (38.47%). In Oregon, when a zip code's fire instances increase from the 25th percentile to the 75th percentile, a 1% increase in a zip code's forest coverage leads to an additional -.44% impact of wildfire on house value. These results can be seen in Appendix Tables 9, 10 and 11.

In addition to a zip code's forest and vegetation coverage, the amount of developed land in a zip code has an effect on the way risk of wildfires is perceived. An increase in the percentage of developed land decreases the impact of wildfires on house value. This may be because in highly developed areas there is a lower perceived threat of nearby wildfires impacting infrastructure, perhaps due to more firefighting efforts in highly developed areas than in rural areas. In less developed zip codes, wildfires may also be closer to houses than in highly developed zip codes. These results can be seen in Appendix Table 12.

When spillover effects are considered, an even smaller effect of wildfires on a zip code's house value is found. An additional fire per square mile in a zip code leads to a -0.003% change in the house value when both least distance and second-least distance fires are considered, compared to a -0.0863% change when second-least distance fires are not considered. This indicates that initial findings, when spillover effects are not considered, may actually be more accurate. As zip codes in the west are very large compared to the size of fires, homes in a zip code that is not the least distance to a fire are unlikely to be affected. When each state is analyzed individually, the spillover effects are also lower than initially found effects. These results are displayed in Appendix Tables 13 and 14, and effects by state are displayed in Appendix Table 15.

Conclusion

In summation, to analyze the effects of wildfires on house value, zip codes may be too large to yield consistent results. As the magnitude of most results is very small compared to results found in previous studies, including Loomis (2004), Mueller & Loomis (2014) and Keil & Matheson (2018), it is likely that there is too much variation of risk within zip codes to see a causal effect between wildfires and change in house value at the zip code level. There is some evidence that wildfires have a greater impact on house price for lower valued homes, as the magnitude of effect is greater when the 25th quantile of house price is analyzed than the 75th quantile. Additionally, the lagged model shows the effect of wildfires on house price is persistent over time.

The percentage of vegetation and forest coverage in a zip code amplifies the effect of wildfires on house price. Forest coverage in particular has a larger effect than general vegetation coverage, which may be due to the view that forests are an amenity, which can be destroyed by wildfires. Upon examination of the percentage of land development in a zip code, there is evidence that areas with low development have a higher impact of wildfires on house values than areas of high development. This may be because houses in rural areas are closer to the actual locations of wildfires, or there is less firefighting infrastructure in place.

Future research should use smaller units of area than zip code in order to capture the precise spatial effects of wildfires on house price. In addition, information on the amount of firefighting infrastructure should be considered, as successful firefighting efforts can mitigate the perceived risk of wildfires, as Baylis and Boomhower (2019) suggest. As this paper does not account for firefighting expenditure, but only the average house value, the impact of increased

wildfires is likely underestimated. As such, this paper finds that the perceived effect of wildfires on house value varies across regions, and is affected by local characteristics.

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